

Supermarket Sales Analysis

Comprehensive Data Analysis Report

Project 1: Retail Domain Analysis

Report Date: September 06, 2026

Dataset: 2,000 Supermarket Transactions

Executive Summary

This report analyzes 2,000 supermarket transactions across three branches and six product categories.

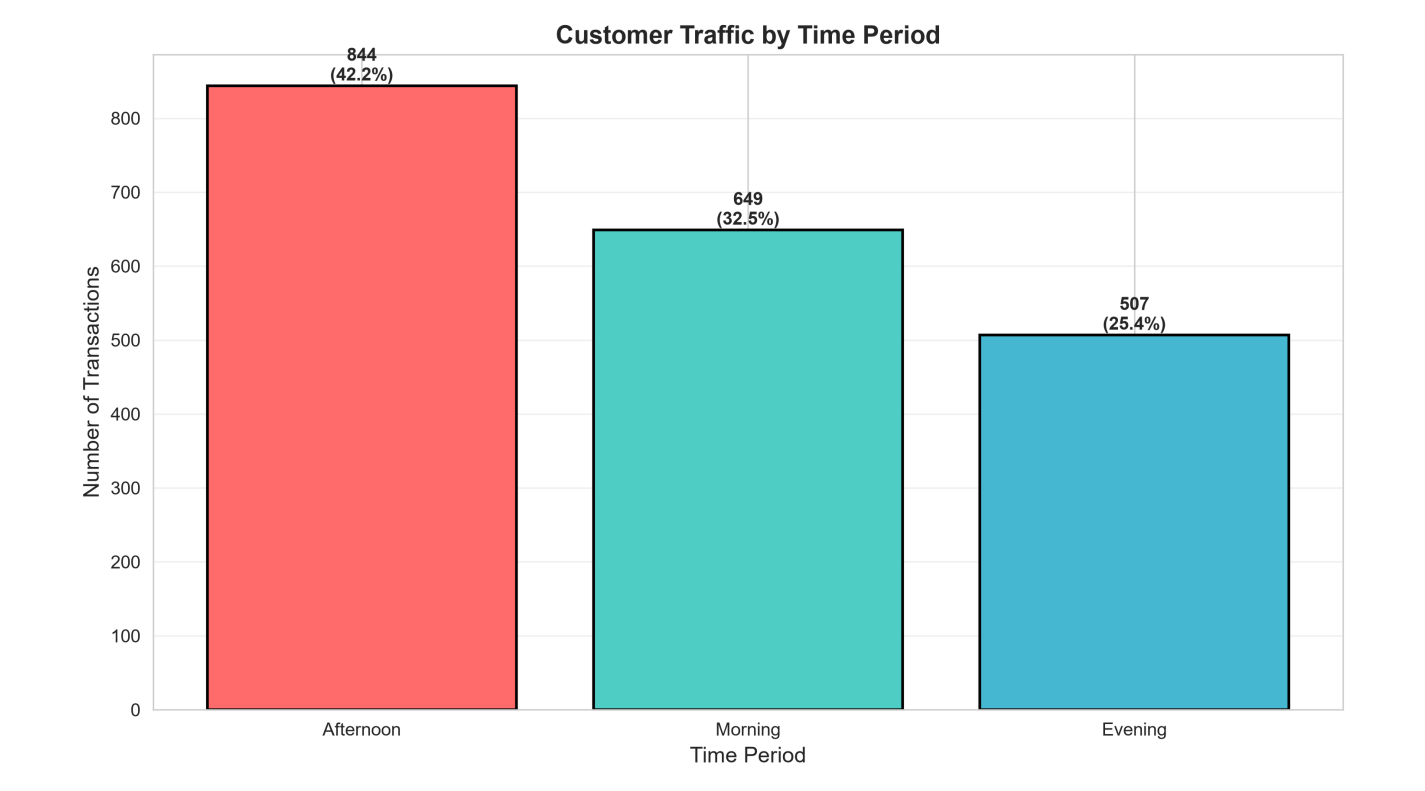
KEY FINDINGS:

- Total Revenue: Rs. 519,280.69
- Average Transaction Value: Rs. 259.64
- Peak Period: Afternoon (42.2% of transactions)
- Best Branch: Branch B (34.8% of revenue)
- Overall Satisfaction: 6.96/10 (Moderate-to-Good)

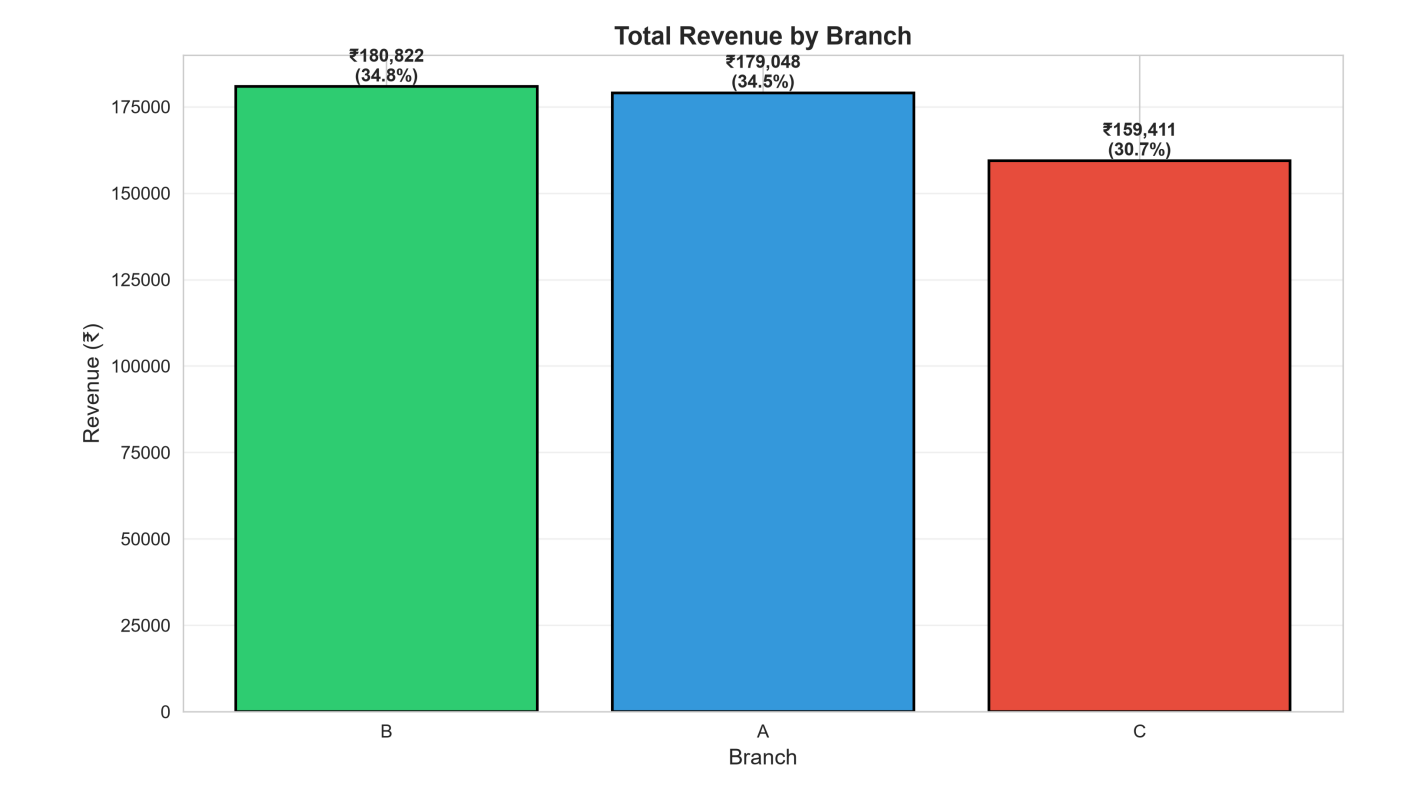
The analysis reveals significant opportunities for operational optimization, particularly in staffing during peak hours and addressing Branch C underperformance.

Key Visualizations

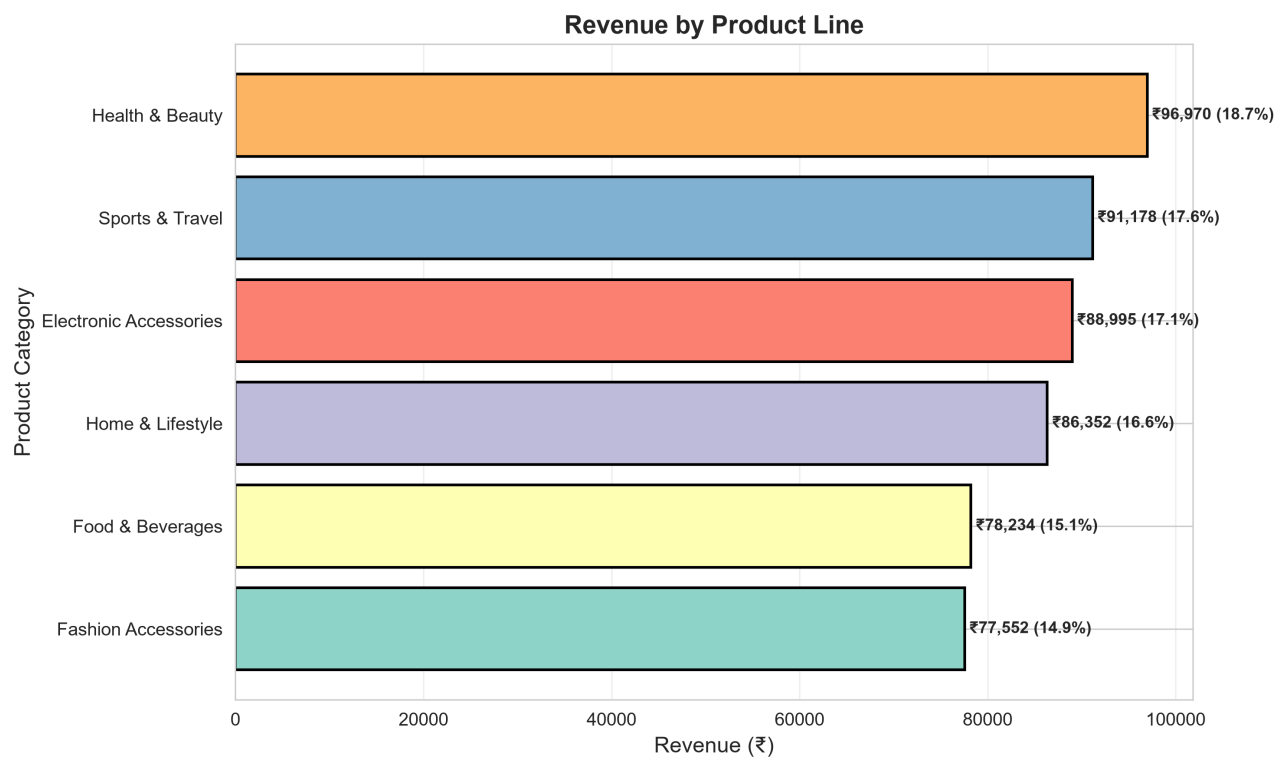
Customer Traffic by Time Period



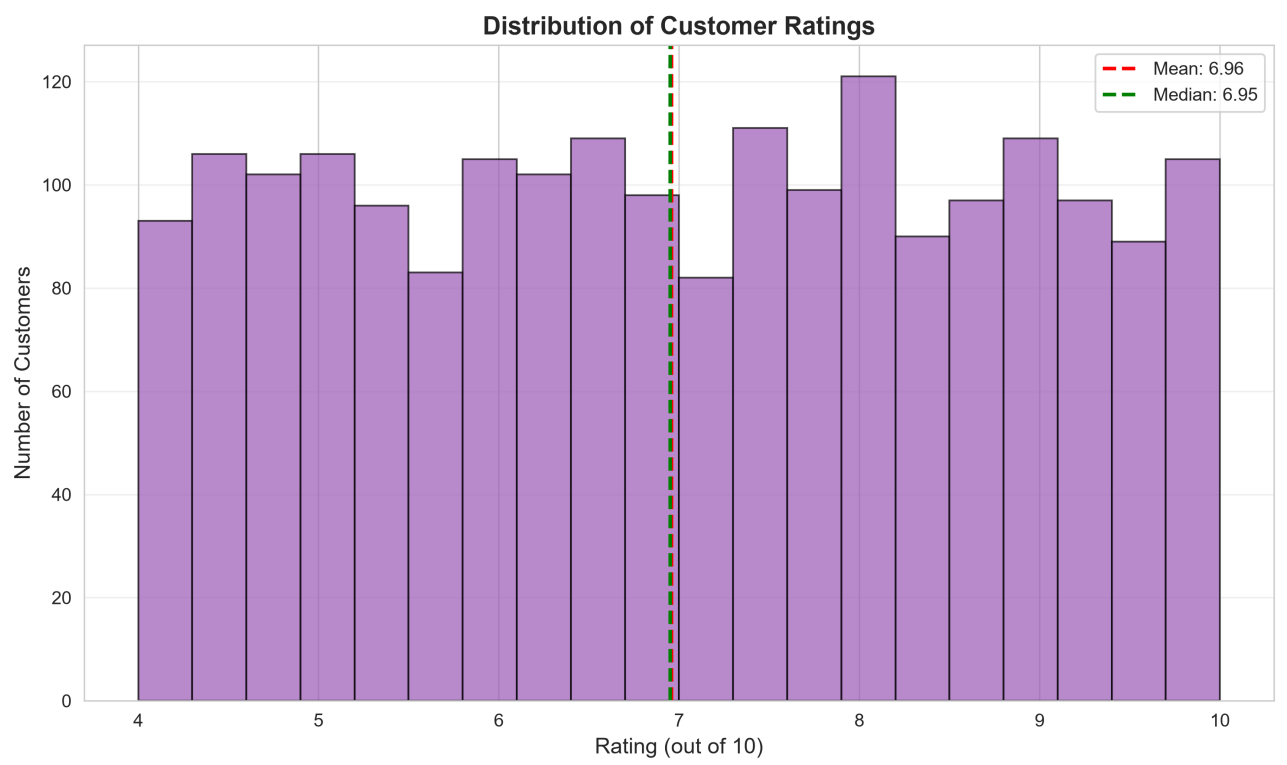
Revenue by Branch



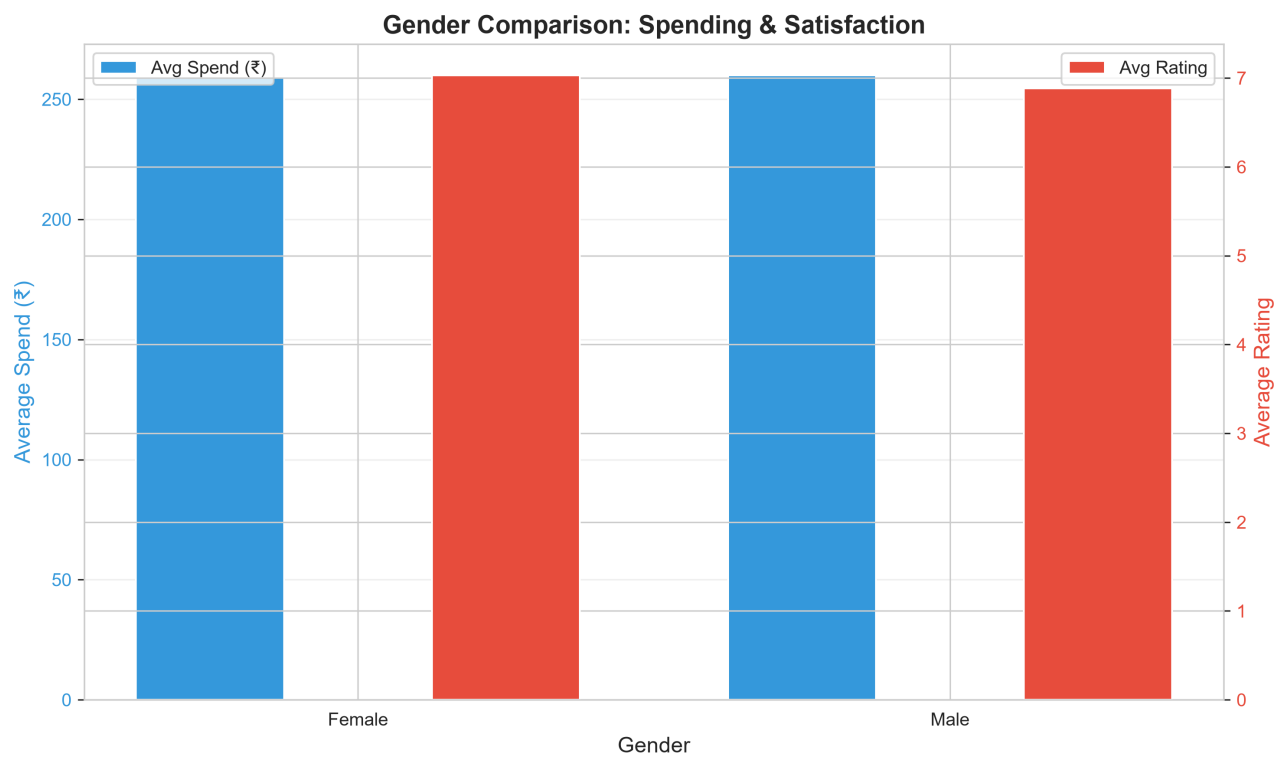
Revenue by Product Line



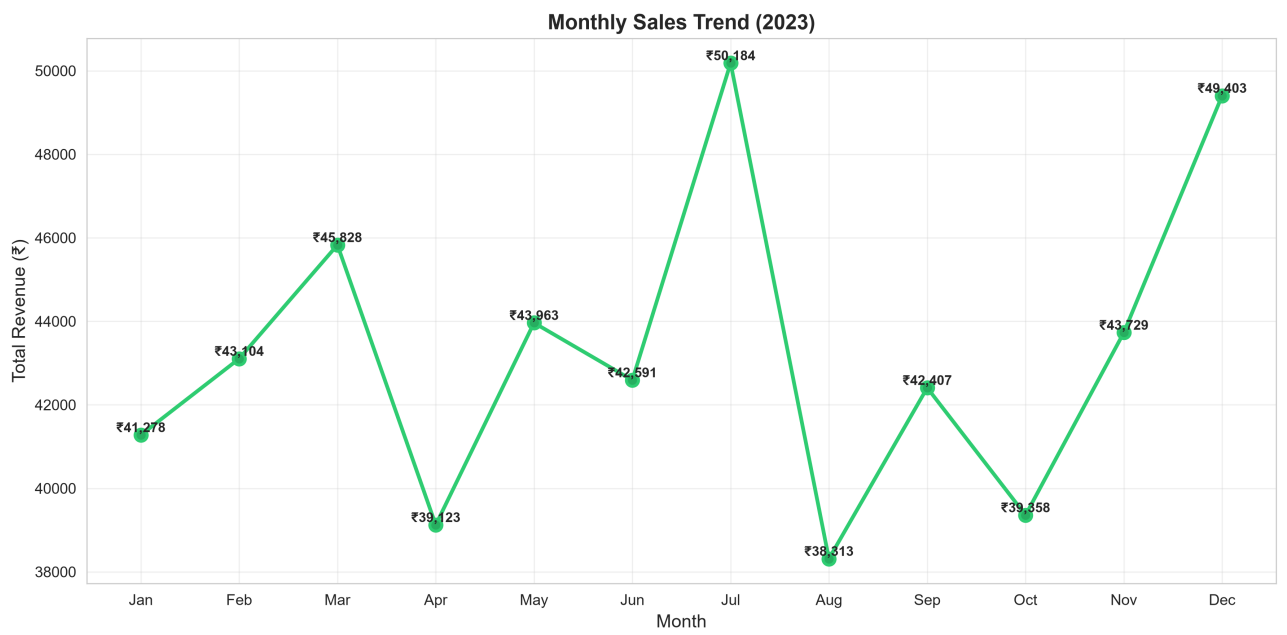
Customer Rating Distribution



Gender Comparison



Monthly Sales Trend



Key Findings

1. OPERATIONAL PATTERNS

Afternoon peak (42%) requires strategic staffing and inventory management.

2. BRANCH PERFORMANCE

Branch C underperforms by 4% compared to Branch B - needs investigation.

3. PRODUCT PORTFOLIO

Well-balanced across categories with Health & Beauty leading at 18.7%.

4. CUSTOMER SATISFACTION

Average rating 6.96/10 indicates moderate-to-good satisfaction with 19% low-rating segment.

5. GENDER DYNAMICS

Female customers generate 3.68% more revenue through higher visit frequency.

6. SEASONAL TRENDS

July peak (Rs. 50,184) followed by August dip - clear seasonal patterns.

Strategic Recommendations

IMMEDIATE ACTIONS:

- Increase afternoon staffing by 20-25%
- Investigate Branch C performance
- Boost evening promotions

MEDIUM-TERM:

- Female-focused marketing campaigns
- Address low-satisfaction segment
- Seasonal promotion planning

EXPECTED OUTCOMES:

- 15-20% operational efficiency improvement
- 10-15% Branch C revenue increase
- 5-10% satisfaction improvement
- 25-30% higher average transaction value